



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic decisions

The Foundation

3.9K

Total Purchases

Transactions analyzed

18

Data Points

Features per customer

4

Categories

Product types tracked

Key Features Captured

- Customer demographics: Age, Gender, Location
- Purchase details: Item, Category, Amount, Season
- Shopping behavior: Discounts, Reviews, Frequency
- Subscription status and shipping preferences

Python Analysis Pipeline

01

Data Loading & Exploration

Imported dataset, checked structure and summary statistics

02

Cleaning & Standardization

Handled 37 missing values, standardized column names

03

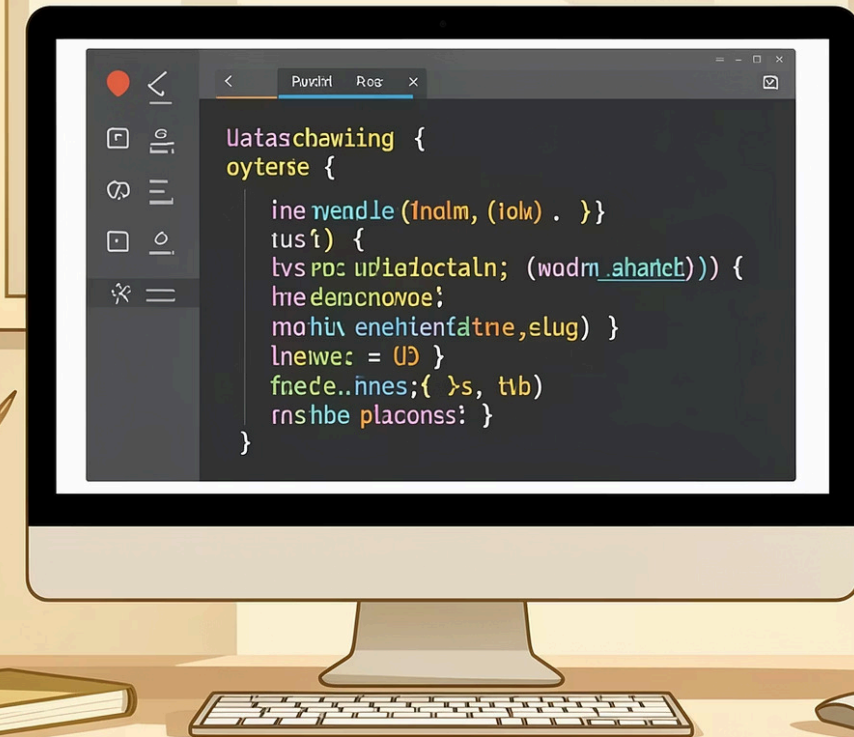
Feature Engineering

Created age groups and purchase frequency metrics

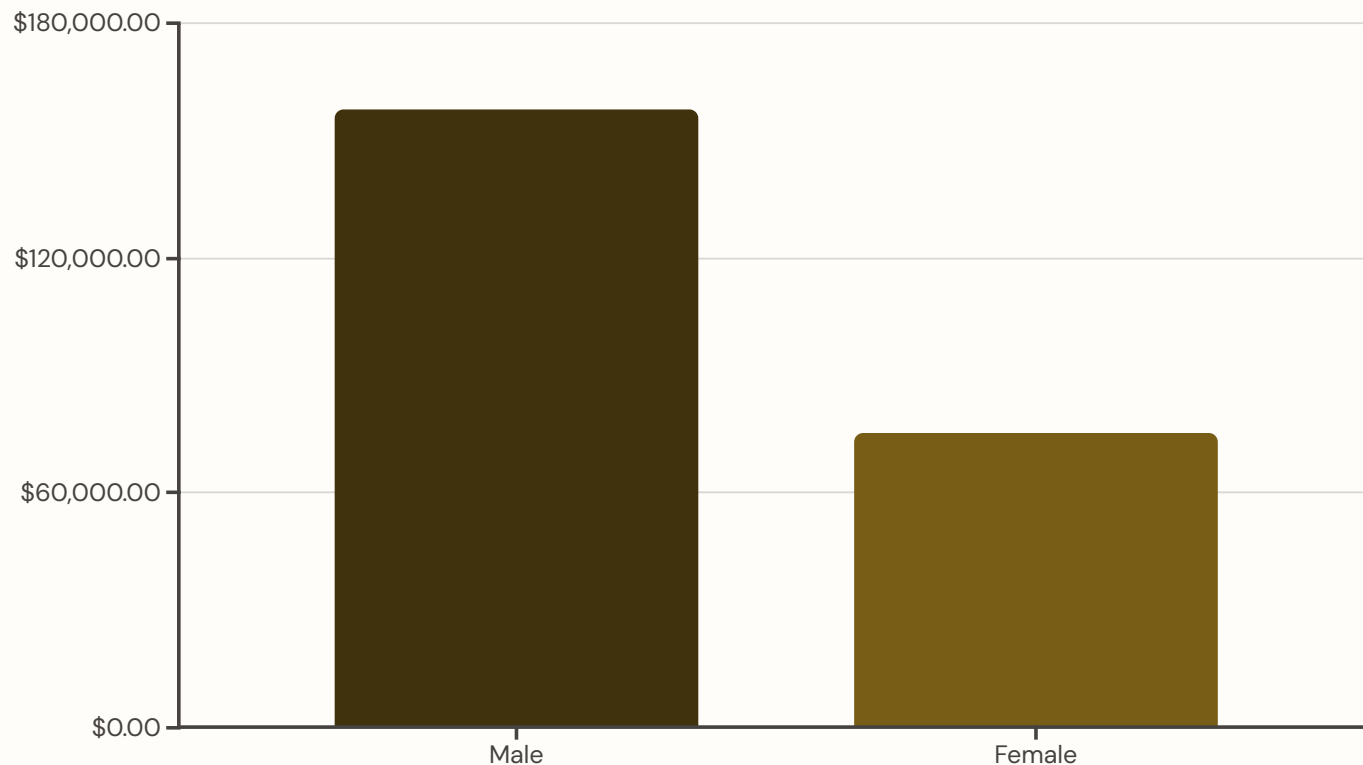
04

Database Integration

Connected to PostgreSQL for advanced SQL analysis



Revenue Patterns Revealed



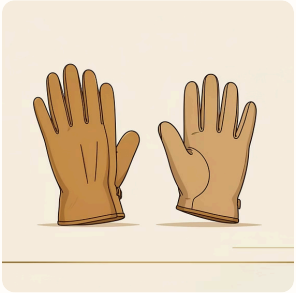
Gender Revenue Gap

Male customers generate **2.1x more revenue** than female customers

Total revenue analyzed: **\$233,081**

Average purchase: **\$59.76**

Top-Rated Products Drive Quality



Gloves

Rating: **3.86**



Sandals

Rating: **3.84**



Boots

Rating: **3.82**



Hat

Rating: **3.80**

Loyalty Distribution

Customer Base Breakdown

Understanding purchase behavior
patterns across 3,900 customers

Loyal

3,116 customers

80% of base

Returning

701 customers

18% of base

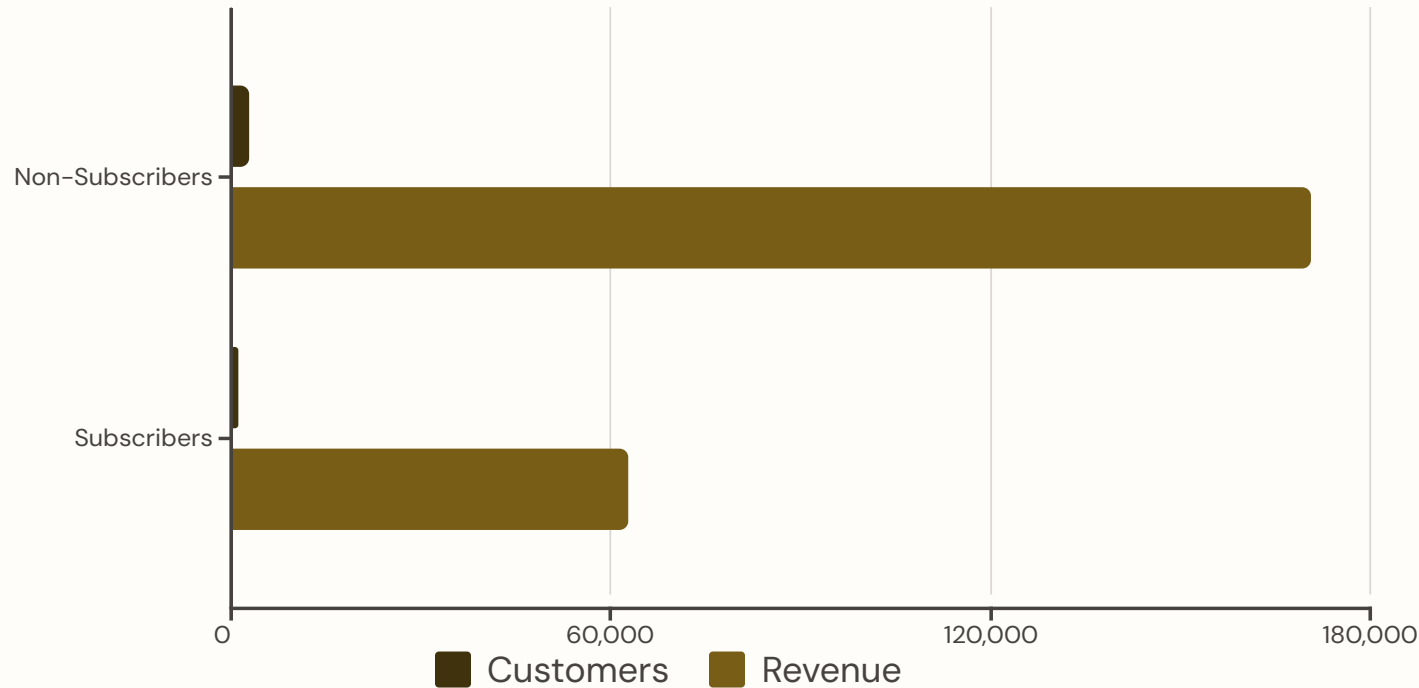
New

83 customers

2% of base



Subscription Impact Analysis



Key Findings

Only **27% subscribe** despite similar spending patterns

- Subscribers: \$59.49 avg spend
- Non-subscribers: \$59.87 avg spend
- Huge growth opportunity in conversion



POWER BI

Interactive Dashboard

Real-Time Metrics

Track customer count, average purchase, and review ratings

Revenue Analysis

Compare performance across categories and age groups

Segment Insights

Filter by subscription status, gender, and shipping type

STRATEGIC RECOMMENDATIONS

Action Plan for Growth



Boost Subscriptions

Promote exclusive benefits to convert
73% non-subscribers



Loyalty Programs

Reward repeat buyers to grow loyal
segment



Targeted Marketing

Focus on high-revenue age groups
and express shipping users

Key Takeaways



Strong Loyalty Base

80% loyal customers provide stable foundation



Subscription Opportunity

27% conversion rate shows massive growth potential



Quality Products

3.75 average rating validates product excellence

